

# Group Project Proposal

## IE 594 — Data Science I

| Item       | Requirement                                                                           |
|------------|---------------------------------------------------------------------------------------|
| Teams      | 1–3 students                                                                          |
| Due        | Friday, October 9, 2026, 11:59 pm                                                     |
| Submission | One PDF per team                                                                      |
| Length     | Maximum 3 pages for the main proposal; references and AI/prompt appendix do not count |
| File name  | IE594_ProjectProposal_TeamName.pdf                                                    |

### 1. Purpose of the Proposal

The proposal should show that your team has identified a meaningful engineering or operations problem, found a suitable real dataset, and designed a defensible analysis plan. The goal is not to predict which algorithm will “win.” The goal is to demonstrate that you understand the decision context, the data-generating process, the risks in the data, and how you will evaluate whether a model is useful.

**Central question** — What decision will your analysis support, and what evidence will convince you that the model is trustworthy enough for that decision?

### 2. Project Scope and Dataset Requirements

- **MIE relevance:** Choose a practical Mechanical or Industrial Engineering problem such as predictive maintenance, quality control, manufacturing, energy, transportation, supply chain, healthcare operations, reliability, safety, or human/AI decision support.
- **Real data:** Use a public, citable dataset with a documented source and license. Restricted datasets are acceptable only if every team member has legitimate access.
- **Target:** Your project must contain a clearly defined regression or classification target tied to a real decision.
- **Scale:** Use a dataset large and rich enough to support model comparison and validation. As a general guideline, aim for at least 1,000 observations when feasible and substantially more observations than predictors.
- **Predictors:** Use enough predictors to require meaningful preprocessing and model selection; there is no fixed 20–30 feature requirement if the project is otherwise technically substantive.
- **No toy projects:** Do not use Iris, Titanic, Boston Housing, MNIST digits, or another heavily recycled classroom dataset unless you obtain instructor approval for a genuinely different research question.
- **Ethics/privacy:** No personally identifiable information. Follow the dataset license and document any relevant privacy, bias, safety, or representativeness concerns.

### 3. Required Proposal Content

#### A. Problem, Stakeholder, and Decision (about 0.5 page)

- Describe the engineering/operations problem in concrete terms.
- Identify the stakeholder or user of the model.
- State the exact target, units/classes, and prediction time or horizon.
- Explain what decision the prediction will inform and what the cost of a wrong decision may be.

#### B. Data Description and Provenance (about 0.5 page)

- Provide the dataset URL/citation, license, collection context, and date range if relevant.
- Report number of rows, predictors, target distribution, data types, and whether observations are grouped, repeated, spatial, or temporal.

- Identify known or suspected quality issues: missingness, outliers, duplicates, censoring, label quality, imbalance, or sampling bias.
- Identify any variables that may create target leakage or would not be available at prediction time.

### C. Proposed Data and Validation Design (about 0.5 page)

- Describe preprocessing that must be learned from data: imputation, encoding, scaling, feature construction, transformations, etc.
- Specify the train/validation/test strategy. Explain why a random, stratified, grouped, or time-aware split is appropriate.
- State explicitly how you will prevent preprocessing and model-selection leakage.

### D. Modeling and Evaluation Plan (about 0.75 page)

- Define a simple baseline: linear regression for regression or logistic regression for classification.
- State the primary evaluation metrics and explain why they reflect the stakeholder decision.
- For classification, address threshold choice and asymmetric errors when relevant. For regression, address the practical meaning of error magnitude.

### E. Project-Specific Analysis Question

Every project must include at least one analysis that cannot be answered by simply calling fit/predict on several models. Choose one project-specific question such as:

- How does performance change under a realistic distribution shift or future-time split?
- Which sensors/features can be removed with little loss in performance?
- How sensitive is the decision to the classification threshold or cost assumptions?
- Does performance or calibration differ across operational subgroups?
- How robust are conclusions to missingness, noisy measurements, or alternate preprocessing choices?
- Which model errors are most operationally consequential, and what patterns characterize them?

### F. Success Criteria, Risks, and Verification Plan

- Define at least one quantitative success criterion relative to a baseline or operational requirement. Avoid arbitrary numbers with no justification.
- List the two or three most important threats to validity and how you will test or mitigate them.
- Describe at least two independent checks you will use to verify the analysis (for example: reproduce a metric manually on a small subset, compare two implementations, inspect residual/error cases, or run a leakage sanity check).

### G. Team Plan and Milestones

- Assign initial responsibilities, but expect every member to understand the complete project.
- List 3–5 milestones from data audit through final evaluation.
- Identify one foreseeable bottleneck and a fallback plan.

## AI Use Policy: Limited Technical Assistance Only

Generative AI may be used only for limited technical assistance. It may help with implementation, but it may not generate the intellectual content of the project. The project is designed to assess your own problem formulation, methodological choices, interpretation, and technical judgment.

| Permitted uses                                                                                                                                                                                                                                             | Not permitted                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Debugging code that the team has written.</li> <li>• Explaining programming errors or library syntax.</li> <li>• Suggesting alternative syntax for implementing a method already selected by the team.</li> </ul> | <ul style="list-style-type: none"> <li>• Selecting the project problem or research question.</li> <li>• Choosing the dataset on behalf of the team.</li> <li>• Performing or interpreting exploratory data analysis.</li> <li>• Selecting predictors, transformations, models, metrics, validation strategies, or thresholds.</li> <li>• Generating explanations of results.</li> </ul> |

|                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>Minor grammar, formatting, or readability edits.</li> </ul>                                                                                                               |                                                                                                                                                                                                                                                                                                                                       |
| <ul style="list-style-type: none"> <li>All AI-assisted technical work must be independently checked.</li> <li>Students remain responsible for correctness and must be able to explain submitted code.</li> </ul> | <ul style="list-style-type: none"> <li>Writing the proposal/report discussion, conclusions, limitations, or recommendations.</li> <li>Deciding which model is preferable or why.</li> <li>Generating answers for the presentation or oral defense.</li> <li>Asking AI to complete an entire analysis pipeline or notebook.</li> </ul> |

Required AI disclosure. If generative AI is used while preparing the proposal, include an AI Use Appendix containing: (1) the exact prompt(s), (2) the AI-generated response or relevant excerpt, (3) what your team used, modified, or rejected, and (4) how the team independently verified the result. If no generative AI was used, state: “No generative AI was used in this project proposal.”

Responsibility for submitted work. Students must understand every line of submitted code and every analytical choice. AI-generated interpretations, methodological choices, conclusions, or recommendations are not permitted. Any team member may be asked to explain code, preprocessing, modeling decisions, outputs, assumptions, or what would happen if a parameter or assumption changed.

## 4. Individual Accountability

**Short proposal check-in** — After proposals are submitted, each team should be prepared for a brief in-class or instructor check-in. Any member may be asked to explain the target, split design, leakage risks, baseline, and one proposed metric without using AI or notes.

## 5. Proposal Rubric — 100 points

| Criterion                             | Points | What strong work demonstrates                                                                  |
|---------------------------------------|--------|------------------------------------------------------------------------------------------------|
| Problem, stakeholder, decision        | 15     | Specific engineering/operations context; target and decision are clear and meaningful.         |
| Data provenance and quality audit     | 15     | Credible source; schema understood; quality, bias, leakage, and availability risks identified. |
| Validation and preprocessing design   | 20     | Split matches data structure; preprocessing is leakage-safe; final test set is protected.      |
| Modeling and evaluation plan          | 20     | Baseline plus well-motivated alternatives; metrics match stakeholder costs and goals.          |
| Project-specific analysis question    | 10     | A substantive question beyond generic model comparison.                                        |
| Success criteria, risks, verification | 10     | Concrete criteria; threats to validity; independent verification plan.                         |
| AI-use plan, team plan, clarity       | 10     | Transparent AI plan; feasible milestones; concise and professional proposal.                   |

## 6. Proposal Template

Title:

Team members and roles:

### A. Problem, Stakeholder, and Decision

[Write your response here.]

### B. Data Description and Provenance

[Write your response here.]

### C. Proposed Data and Validation Design

[Write your response here.]

**D. Modeling and Evaluation Plan**

[Write your response here.]

**E. Project-Specific Analysis Question**

[Write your response here.]

**F. Success Criteria, Risks, and Verification Plan**

[Write your response here.]

**G. Team Plan and Milestones**

[Write your response here.]

**H. Planned AI Use & Verification**

[Write your response here.]

References

Appendix: AI Use Appendix if applicable (exact prompts, relevant AI output, what was used/modified/rejected, and independent verification); optional supporting figures/tables.